

What is a Lookup Function?

A **lookup function** searches for a value in a table and returns related information.

Real-life Example

Imagine a school.

Instead of searching through thousands of student records manually, you search using a **Student ID**.

Student ID → 102

Excel finds:

- Name → Priya
- Class → 10
- Marks → 95

This is exactly what lookup functions do.

Sample Data

A (ID) B (Name) C (Department) D (Salary)

101	Ravi	HR	30000
102	Priya	IT	50000
103	Arun	Finance	45000
104	Neha	IT	60000
105	Kiran	Sales	40000

Suppose you only know:

Employee ID = 104

You want to find:

- Name
- Department
- Salary

Lookup functions do this automatically.

Types of Lookup Functions

The most common lookup functions are:

1. **VLOOKUP()**

2. **HLOOKUP()**
 3. **XLOOKUP()**
 4. **LOOKUP()** (older function, used less often)
 5. **INDEX() + MATCH()** (advanced and very useful)
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1. VLOOKUP()

V stands for **Vertical**.

It searches **vertically** (top to bottom) in the **first column** of a table.

Syntax

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Understanding Each Argument

1. lookup_value

The value you want to search.

Example:

104

2. table_array

The entire table where Excel should search.

Example:

A2:D6

3. col_index_num

The column number (counting from the left of the selected table).

Example:

Column	Number
ID	1
Name	2
Department	3

Column	Number
--------	--------

Salary	4
--------	---

4. range_lookup

- FALSE → Exact match (most commonly used)
- TRUE → Approximate match (used only with sorted data)

For Data Analyst work, always use **FALSE** unless you specifically need approximate matching.

Example 1 – Find Employee Name

Data:

ID	Name
----	------

101	Ravi
-----	------

102	Priya
-----	-------

103	Arun
-----	------

Formula:

=VLOOKUP(102,A2:B4,2,FALSE)

How Excel Works

Step 1:

Search for **102** in the first column.

↓

Find row:

102 | Priya

↓

Column number = 2

↓

Return

Priya

Example 2 – Find Salary

Formula:

=VLOOKUP(104,A2:D6,4,FALSE)

Excel:

Search ID 104

↓

Go to column 4

↓

Return

60000

Why FALSE is Important

Using:

=VLOOKUP(104,A2:D6,4,FALSE)

means Excel must find **exactly 104**.

If 104 does not exist:

Result:

#N/A

Example with Cell Reference

Suppose:

F2 contains:

103

Formula:

=VLOOKUP(F2,A2:D6,2,FALSE)

Result:

Arun

Now change F2 to:

104

The result automatically changes to:

Neha

Limitations of VLOOKUP()

1. Searches only from left to right

Example:

Name ID

Ravi 101

Priya 102

You cannot search for **Name** and return **ID** using VLOOKUP because the lookup value is not in the first column.

2. Column numbers can break

Suppose:

|ID|Name|Department|

You use:

=VLOOKUP(102,A:C,2,FALSE)

If you insert a new column between ID and Name, the column numbers change and the formula may return the wrong result.

2. HLOOKUP()

H stands for **Horizontal**.

It searches across the **first row**.

Example:

	A	B	C
--	---	---	---

1 ID	101	102
------	-----	-----

2 Name	Ravi	Priya
--------	------	-------

Formula:

=HLOOKUP(101,A1:C2,2,FALSE)

Result:

Ravi

3. XLOOKUP()

This is the newest and most powerful lookup function in Excel.

It is easier to write and more flexible than VLOOKUP.

Syntax

=XLOOKUP(lookup_value, lookup_array, return_array)

Example

ID Name

101 Ravi

102 Priya

103 Arun

Formula:

=XLOOKUP(102,A2:A4,B2:B4)

Result:

Priya

Another Example

Return salary.

=XLOOKUP(104,A2:A6,D2:D6)

Result:

60000

Advantages of XLOOKUP()

Can search left or right

Example:

Name ID

Ravi 101

Priya 102

Formula:

=XLOOKUP("Priya",A2:A3,B2:B3)

Result:

102

VLOOKUP cannot do this.

No column numbers

Instead of:

Column 4

You simply tell Excel which column to return.

This makes formulas easier to read and less likely to break.

Better error handling

You can display your own message if nothing is found.

Example:

```
=XLOOKUP(120,A2:A6,D2:D6,"Employee Not Found")
```

If ID 120 doesn't exist:

Result:

Employee Not Found

Instead of:

#N/A

4. LOOKUP()

Older function.

Example:

```
=LOOKUP(102,A2:A6,B2:B6)
```

It is still available but is rarely used in modern Excel because XLOOKUP is more powerful.

5. INDEX() + MATCH()

This combination is widely used in professional Excel models.

MATCH()

Finds the position of a value.

Example:

```
=MATCH(103,A2:A6,0)
```

Result:

because 103 is the third item in the selected range.

INDEX()

Returns a value from a given position.

Example:

=INDEX(B2:B6,3)

Result:

Arun

Combine INDEX and MATCH

=INDEX(B2:B6,MATCH(103,A2:A6,0))

How it works:

MATCH finds the position of 103 (3)

↓

INDEX returns the 3rd value from the Name column

↓

Result:

Arun

VLOOKUP vs XLOOKUP

Feature	VLOOKUP	XLOOKUP
Search direction	Left to right only	Left or right
Uses column numbers	Yes	No
Handles missing values	Returns #N/A unless wrapped in another function	Can return a custom message
Easy to read	Moderate	Yes
Recommended for new Excel versions	Good	Best choice

Real Data Analyst Example

Suppose you have:

Employee Table

Employee ID	Name	Department	Salary
-------------	------	------------	--------

101	Ravi	HR	30000
102	Priya	IT	50000
103	Arun	Finance	45000

Task

You receive only an Employee ID:

102

You need to prepare a report showing:

- Employee Name
- Department
- Salary

Using lookup functions:

- XLOOKUP() or VLOOKUP() finds the employee's details instantly.
- This is common when combining HR, sales, inventory, or customer data from different worksheets.

Interview Questions

1. What is a lookup function?

A lookup function searches for a value in a table and returns related information.

2. What is the difference between VLOOKUP and XLOOKUP?

- **VLOOKUP** searches only from left to right and uses column numbers.
- **XLOOKUP** can search in any direction, doesn't require column numbers, and can return a custom message if no match is found.

3. Why is FALSE commonly used in VLOOKUP?

Because it tells Excel to return only an **exact match**, which is what you usually need in business and data analysis.

4. When would you use INDEX + MATCH instead of VLOOKUP?

When you need more flexibility, such as looking up values to the left, or when you want formulas that are less likely to break if columns are inserted or moved.

